FFGFT in the RA 2.1 Architecture

A One-Page Diagnostic Entry

Fundamental Fractal Geometric Field Theory
mapped onto the Representational Architecture (Guevara 2026)

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List of Symbols

Symbol	Meaning
$\xi = 4/30000$	FFGFT dimensionless winding parameter; geometrically fixed via $\kappa = 7$ (not fitted)
T^4	4-dimensional torus $(\mathbb{R}/2\pi\mathbb{Z})^4$; FFGFT ontological primitive
$n_\theta, n_\phi, n_3, n_4$	Integer winding numbers in the four torus directions
θ_4	Continuous fourth-phase coordinate on T^4
$\varepsilon \approx 0.001$	Non-closure of T^4; generative structural residual
$\tilde{T} \cdot m = 1$	Foundational T0 relation (time-mass duality)
$D_f = 3 - \xi$	Fractal dimension of physical space (≈ 2.99987)
$\kappa = 7$	Unique winding exponent from three measured mass ratios
$L^2(T^4)$	FFGFT Hilbert space; square-integrable functions on T^4
α	Fine structure constant; derived from ξ (Doc 009/137)
Λ_{FFGFT}	Cosmological constant: 1.34×10^{-122} Planck units (Doc 182)
H_0	Hubble constant from ξ: 66.2 km/s/Mpc, -1.9% from Planck (Doc 182)
$\tau \cdot \mu$	Coherence-time–information-mass product $= \hbar/c^2$; exact
RA4–RA1	Layers of the Representational Architecture, introduced by J.T. Guevara Calderón (2026), *RA 2.1*, IPI working document, May 2026

FFGFT Diagnostic Entry — RA 2.1 Format

Fundamental Fractal Geometric Field Theory (T0 / Time-Mass Duality)

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RA nomenclature: Guevara Calderón, J.T. (2026). *Representational Architecture 2.1.* IPI working document, May 2026.

Layer	FFGFT Content	Standard Model / ΛCDM	RA Status
RA4 Ontol.	4D torus T^4 with winding structure $(n_\theta, n_\phi, n_3, n_4)$, continuous phase θ_4, and non-closure $\varepsilon \approx 0.001$. Single parameter $\xi = 4/30000$ geometrically fixed from three measured mass ratios via $\kappa = 7$. Foundational relation $\tilde{T} \cdot m = 1$.	Quantum fields on Minkowski spacetime $\mathbb{R}^{3,1}$. Multiple independent coupling constants, masses, and mixing angles as inputs. Λ identified with QFT vacuum energy (Zeldovich step 3).	**RA4 divergence.** Primitive replaced. Zeldovich step 3 not made; no tuning problem.
RA3			

continued...

Layer	FFGFT Content	Standard Model / ΛCDM	RA Status
Admiss.	Only winding configurations consistent with ξ are physically admissible. Fractal dimension $D_f = 3 - \xi \approx 2.99987$ defines accessible topological sectors. Z3^3 topological constraint restricts to exactly three particle generations. No additional axioms needed.	All renormalizable field configurations admissible. Particle content and number of generations are experimental inputs, not derived from any admissibility constraint.	**RA3 restriction.** ξ maximally constrains admissibility. SM leaves it open; generations unexplained.
RA2			
Repres.	$L^2(T^4)$ as state space. Fourier modes with winding numbers $\mathbf{n} \in \mathbb{Z}^4$ as natural basis. Standard QM operator algebra and Born rule emerge as projections from torus geometry (Doc 230/231, Hilbert space bridge). No separate quantization postulate.	Fock space over Minkowski spacetime. Operator algebra and Born rule postulated independently of spacetime structure. Quantization is an additional input.	**RA2 derivation.** QM formalism derived from T^4 geometry. In SM it is postulated independently.
RA1.5			
Operat.	All derived from ξ alone, **no fitting**: • Lepton and quark masses via $\kappa = 7$ recursion; agreement $< 1\%$ • Fine structure constant α (Doc 009/137) • CMB temperature; deviation $\Delta = 0.0002\%$ (Doc 025) • $\Lambda_{\text{FFGFT}} = 1.34 \times 10^{-122}$ Planck units (Doc 182) • $H_0 = 66.2$ km/s/Mpc (-1.9% from Planck) • $\tau \cdot \mu = \hbar/c^2$ (exact)	All quantities above are independent experimental inputs requiring separate measurement. Λ requires 120-order fine-tuning relative to QFT zero-point energy.	**RA1.5 unification.** All SM constants from one parameter. SM treats them as unrelated experimental inputs.
RA1			
Expr.	Concrete numerical values for all Standard Model constants and cosmological parameters, derived not fitted. Published in 149+ documents; GitHub: jpascher/T0-Time-Mass-Duality; Zenodo v1.1.0, DOI 10.5281/zenodo.20117635.	Parameter values measured and inserted by hand. No unifying derivation. SM has ≈ 20 free parameters; ΛCDM adds several more.	**RA1 compression.** 1 parameter vs. $\sim$20 (SM).

Falsification criteria (RA1 level):

- Any lepton or quark mass ratio inconsistent with the ξ-ladder at $\kappa = 7$ (tolerance $< 1\%$).
- CMB deviation exceeding the correction term $(275/4) \cdot \xi \approx 9.2 \times 10^{-3}$ (Doc 025).

- A cosmological constant value outside the range derived in Doc 182.
- Any winding quantum not assignable to an integer in $\mathbb{Z}^4$.

Key references: Doc 009 (ξ, $\kappa = 7$) · Doc 013 (SI units) · Doc 025 (CMB) · Doc 182 (Λ, H_0) · Doc 230/231 (Hilbert space) · GitHub · Zenodo v1.1.0